Economics
Higher level
Paper 3

SME Predicted Paper – Moderate

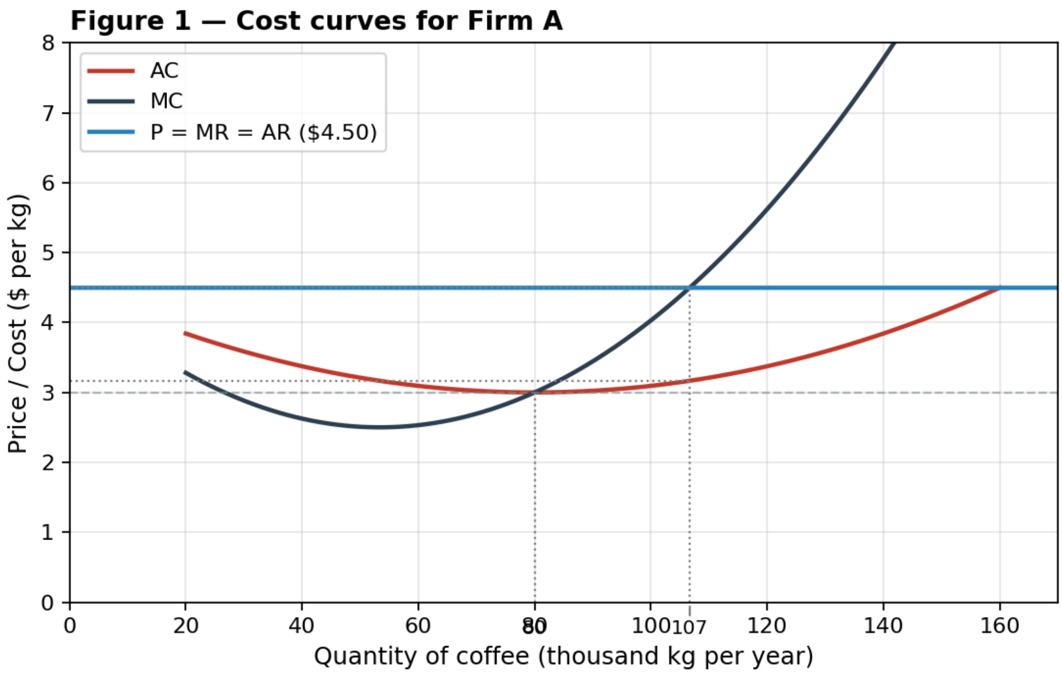
1 hour

Instructions to candidates

- You are permitted access to a calculator for this paper.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers must be given exactly or correct to two decimal places.
- You must show all your working.
- Use fully labelled diagrams where appropriate.
- The maximum mark for this examination paper is [60 marks].

1. Coffee is one of the most widely traded agricultural commodities in the world. Brazil is the largest producer and exporter of coffee, accounting for around 35% of global output.

The global coffee market operates under conditions close to perfect competition at the farm level, with many small producers and a homogeneous product. **Figure 1** below shows the cost curves of Firm A, a typical Brazilian coffee farmer. The current world market price of coffee is $4.50 per kg.



(a) (i) Define the term **price taker**. [2]

(ii) Using **Figure 1**, identify Firm A's short-run profit maximising level of output. [1]

(iii) Using **Figure 1**, calculate Firm A's short-run economic profit/loss at the level of output identified in part (a)(ii). [2]

(iv) With reference to **Figure 1**, identify the long-run equilibrium price and level of output for Firm A. [2]

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(v) Explain, using **Figure 1**, how Firm A will move from short-run equilibrium to long-run equilibrium. [4]

Coffee is a water-intensive crop. According to the WWF it takes approximately 140 litres of water to produce the coffee beans for a single cup. The rapid expansion of coffee production in Brazil contributes to the strain on water resources and is thus responsible for a negative impact on watersheds and local ecosystems.

(vi) State **one** of the characteristics of common access resources. [1]

Still, there are countries that heavily subsidise their coffee farmers so that they are able to meet high global demand. Brazil operates a price-support scheme that guarantees a minimum price of $4.00 per kg. Assume the Brazilian government had to purchase 400 million kg at $4.00 per kg in 2023 to maintain the price floor.

(vii) Using the information above, calculate the total cost of the price-support scheme to the Brazilian government in 2023. [2]

Overall, the coffee industry faces sustainability challenges:

- Coffee production is associated with significant deforestation in tropical regions, reducing forest cover and biodiversity.
- The use of fertilisers and pesticides can contaminate water sources, harm ecosystems and pose risks to human health.
- Coffee processing and transport generate greenhouse gas emissions; reliance on long supply chains contributes to climate change.

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(viii) Using an externalities diagram and the information provided, explain how coffee production is responsible for the generation of negative externalities. [4]

Currently, indirect taxes targeted specifically at coffee-related environmental harms do not exist in Brazil. Standard value-added and consumption taxes (PIS/COFINS, ICMS) do apply. For instance, the average monthly spending per household on coffee in Brazil is R$52.00 while the combined indirect-tax rate on packaged coffee is 25%.

(ix) Using the information above, calculate the indirect tax paid per household each month on coffee in Brazil. [2]

(b) Using the text/data provided and your knowledge of economics, recommend a policy that could be introduced to promote sustainability in the Brazilian coffee industry. [10]

2. Ethiopia is a low-income country in East Africa with a population of around 123 million. Table 1 below provides selected data for Ethiopia.

	Nominal GDP (billion $)	GDP Deflator	Real GDP (billion $)	Population (million)
2020	107.50	100.00	107.50	117.2
2021	111.30	105.30		120.0

Table 1: Selected macroeconomic data for Ethiopia, 2020–2021

(a) (i) Using the information in **Table 1**, calculate the real GDP for Ethiopia in 2021. [2]

(ii) Using the information in **Table 1**, calculate the real GDP per capita for Ethiopia in 2021. [2]

In 2021 Ethiopia held a low position by nominal GDP while its Human Development Index (HDI) ranking placed Ethiopia at 175th out of 191 countries published.

(iii) State **two** indicators that are used to construct the Human Development Index (HDI). [2]

(iv) Outline **one** disadvantage of using the Human Development Index as a measure of economic development. [2]

Ethiopia belongs to the low-income transition economies. Over the past two decades the country has seen rapid growth driven by market-oriented reforms, agricultural-sector expansion and significant foreign direct investment (FDI), particularly into industrial parks that produce textiles and apparel.

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(v) Explain **two** possible consequences of economic growth for Ethiopia. [4]

Table 2 below provides selected data from Ethiopia's balance of payments for 2021 (US$ million).

Item	$ million
Exports of goods	3,625
Imports of goods	14,198
Exports of services	4,350
Imports of services	5,700
Net primary income	-652
Net secondary income	5,860

Table 2: Balance-of-payments data for Ethiopia, 2021

(vi) Using **Table 2**, calculate the current account balance for Ethiopia in 2021. [2]

(vii) Explain **two** implications of a rising current account deficit. [4]

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Ethiopia's export base is narrow. Coffee alone accounts for around 30% of goods exports, followed by oil seeds (10%) and cut flowers (9%). Over the last five years, GDP growth has slowed from around 10% in the 2010s to below 6%, highlighting the vulnerabilities of an economy dependent on primary commodities and with stagnant productivity growth.

(viii) Outline why dependence on primary sector products can pose a barrier to economic growth and/or economic development. [2]

(b) Using the text/data provided and your knowledge of economics, recommend a policy that could be introduced to revitalize economic growth in Ethiopia. [10]